

Emergent fine-structure constant α in Relator theory

One-page conceptual map (two routes, one lock)

1) Gauge-invariant phase rates (what ∇S and $\partial_t S$ mean). Write the state in polar form $\psi = R e^{iS/\hbar}$, where R is interpreted as a *generator-circle radius* in the internal space $\mathbb{C}$ (a length scale attached to the state), while $S(x, t)$ is the usual phase action on $\mathbb{R}^3$.

Electromagnetism enters through the standard gauge freedom

$$\mathbf{A} \mapsto \mathbf{A} + \nabla \chi, \quad \Phi \mapsto \Phi - \frac{1}{c} \partial_t \chi, \quad S \mapsto S + \frac{q}{c} \chi,$$

so the *gauge-invariant* combinations are

$$\mathbf{P}_{\text{mech}} \equiv \nabla S - \frac{q}{c} \mathbf{A}, \quad \mathcal{E}_{\text{GI}} \equiv -\partial_t S - q\Phi.$$

Here ∇S is the ordinary spatial gradient on $\mathbb{R}^3$, and $\partial_t S$ is the ordinary time derivative.

Relator theory turns these invariants into two *rates* (frequencies) in the split space $\mathbb{C} \oplus \mathbb{R}^3$:

$$\omega_{\mathbb{C}} = \frac{\mathcal{E}_{\text{GI}}}{\hbar} = \frac{-\partial_t S - q\Phi}{\hbar}, \quad \omega_{\mathbb{R}^3} = \frac{\|\mathbf{P}_{\text{mech}}\|}{mR} = \frac{\|\nabla S - \frac{q}{c} \mathbf{A}\|}{mR}. \quad (1)$$

2) The Relator postulate and the invariant that drives everything. The foundational kinematic lock is a saturated “phase-speed budget” in $\mathbb{C} \oplus \mathbb{R}^3$:

$$R\omega = c, \quad \omega^2 = \omega_{\mathbb{C}}^2 + \omega_{\mathbb{R}^3}^2. \quad (2)$$

Because $R\omega = c$, ratios remove the length scale R and yield a *pure, dimensionless* gauge-invariant quotient:

$$\Omega_{\mathbb{R}^3} \equiv \frac{\omega_{\mathbb{R}^3}}{\omega} = \frac{\|\nabla S - \frac{q}{c} \mathbf{A}\|}{mc}, \quad D \equiv \Omega_{\mathbb{R}^3}^2. \quad (3)$$

In words: D measures how much of the total phase evolution is spent “winding” through $\mathbb{R}^3$ rather than rotating in $\mathbb{C}$. The internal-clock fraction is

$$\frac{\omega_{\mathbb{C}}}{\omega} = \sqrt{1 - D}.$$

3) What α describes in this theory. α is not inserted as $e^2/(\hbar c)$. It emerges as the *unique dimensionless fixed point* that makes a charged lepton’s *self-field dressing* consistent at a gauge-invariant *matching shell* $r = r_*$, where interior “source primitives” in $\mathbb{C}$ are matched to exterior fields in $\mathbb{R}^3$. On the shell, the slowdown invariant is organized into a scalar–vector decomposition,

$$D = D_{\text{C}}(\alpha) + \Delta D_{\Lambda}(\alpha) + O(\Delta D_{\Lambda}^2), \quad (4)$$

where D_{C} is the Coulombic (scalar) baseline from the phase-gradient sector, and ΔD_{Λ} is the inductive (vector) correction carried by the Λ -channel (a closed, dimensionless inductive sum built from the loop core, UV→IR coarse-grain, exterior subtraction, and shell synchronization).

A useful reduced control parameter is the dimensionless overlap

$$\zeta(\alpha) = \Upsilon_{\text{sp}} \Lambda(\alpha), \quad \Upsilon_{\text{sp}} = \frac{K}{2\pi^2},$$

with K a geometry-only spectral constant.

4) The “two-path” breakthrough: the same log coefficient computed twice. A universal single-log coefficient $C_{\log}$ appears in the long (logarithmic) piece generated on the shell; the key move is that $C_{\log}$ can be obtained by two *independent* arguments and then *locked*.

Path I (loop polarization / Λ -channel). The shell construction yields

$$C_{\log}^{(I)}(\alpha) = \frac{\pi^2}{D_{\text{C}}(\alpha)} \zeta(\alpha) (1 + \zeta(\alpha)). \quad (5)$$

Path II (pure shell TT geometry). A logarithmic annulus has no preferred in-plane transverse–traceless direction. The rank-1 in-plane TT selection contributes a factor 1/2, and spherical isotropy gives $\langle \hat{n}_i \hat{n}_j \rangle_{S^2} = (1/3) \delta_{ij}$, hence $\langle P_T \rangle_{S^2} = (2/3)I$. Together,

$$C_{\log}^{(II)} = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}. \quad (6)$$

So the numerical 1/3 is not a fit parameter: it is a geometric average.

5) Alpha Lock Point (ALP): what “ α comes from” in one sentence. The emergent coupling is the value that closes the shell consistently:

$$C_{\log}^{(I)}(\alpha) = C_{\log}^{(II)} = \frac{1}{3}. \quad (7)$$

This is an “impedance-matching” picture: the Coulombic phase-gradient drive (D_{C}) and the transverse inductive response (through ζ and Λ) must meet on the shell with a universal TT acceptance of 1/3. Only one α satisfies that closure.

Solving the closed root yields the forward prediction (*not as final prediction yet until simulate from another independent path–least action integral or QED like way*)

$$\alpha_{\text{pred}} \approx 0.007297352564326775, \quad \alpha_{\text{pred}}^{-1} \approx 137.0359991770873.$$

6) Same invariant $D \Rightarrow$ a direct handle on $g-2$. In the Relator framework the lepton gyromagnetic factor is an evolution-rate identity:

$$g = \frac{2}{\sqrt{1 - D}}, \quad a_e \equiv \frac{g - 2}{2}. \quad (8)$$

Once α is fixed by the lock (7), the same shell geometry fixes $D_{\text{C}}(\alpha)$ and the Λ -channel contribution in (4), hence fixes D and produces a definite $g-2$ shift through (8).

Minimal takeaway. In Relator theory, α is the dimensionless equilibrium point of a *self-consistency condition* on the matching shell: it is the number that makes the scalar (Coulombic) and transverse (inductive) self-field dressings compatible, while the total phase evolution in $\mathbb{C} \oplus \mathbb{R}^3$ remains “luminal” ($R\omega = c$) and gauge invariant.

References

- [1] M. Pajuhaan (2025). *Alpha*. Zenodo. DOI: 10.5281/zenodo.16944532.